

Lecture “Stereochemistry”

Take-home messages 4

9. Diastereoselective additions to carbonyl compounds

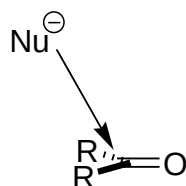
Felkin-Anh-Model:

- Allows to predict which side of acyclic carbonyl groups is attacked if a stereocenter is present in neighbouring position



Bürgi-Dunitz study:

- Nucleophiles attack carbonyl groups along a specific path, the Bürgi-Dunitz trajectory in an angle of 107° .



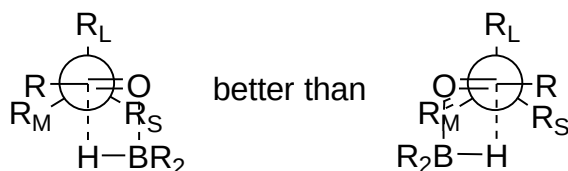
Extended Felkin-Anh model:

- Includes modifications to explain a broader scope of Reactions



1. Conformer leads to fully staggered product (Houk's rule)
2. Minimize Nu – R_S interaction instead of Nu – R_M repulsion
3. Nu attacks along Bürgi-Dunitz trajectory

Internal vs. external Nu-delivery: - Important, when nucleophile and Lewis acid are linked covalently (i.e. boranes).



Corollary to Felkin-Anh model:

„When a polar, electronegative substituent is present a stereoelectronic effect can become dominant over steric effects. In such cases the polar substituent is placed in the position normally reserved for R_L ”

“Normal Felkin-Anh versus Chelate-Felkin-Anh: - If substituents with donor atoms are present (and sterically accessible)
- Chelate case also works for donor atoms in β -, γ -position etc.

